MILITARY SPECIFICATION

FRAME, SHIPPING (SIXCON)

This specification is approved for use by the U.S. Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the SIXCON Shipping Frame.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specification, standards and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	-Preservation, Methods of
MIL-I-6866	-Inspection, Penetrant Method of
MIL-I-6868	-Inspection Process, Magnetic Particle
MIL-W-8611	-Welding, Metal Arc and Gas, Steels, and Corrosion and Heat Resistant Alloys, Process for
MIL-T-27260	-Tiedown, Cargo, Aircraft
MIL-N-27444	-Net, Cargo Tiedown, Pallet
MIL-E-52798	-Enamel, Alkyd, Camouflage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Code LMA-1, Washington, D.C. 20380, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8145

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STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-Marking for Shipment and Storage
MIL-STD-130	-Identification Marking of U. S. Military Property
MIL-STD-453	-Inspection, Radiographic
MIL-STD-794	-Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	-Environmental Test Methods and Engineering Guidelines

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

U.S. Marine Corps

855001B0000	-SIXCON Shipping Frame

CODE OF FEDERAL REGULATIONS (CFR)

49CFR451	-Testing and Approval of Cargo Containers

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted should be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ISO 668	-Freight Containers - External Dimensions and Ratings
ISO 1161	-Series 1 Freight Containers -Corner Fittings -Specification
ISO 1496/I	-Series 1 Freight Containers - Specification and Testing - Part I: General Cargo Containers

(Applications for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10012.)

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1	-Structural Welding Code, Section 5, Qualification - Welding Inspection Manual

(Applications for copies should be addressed to the American Welding Society, 2501 North West 7th Street, Miami Florida 33125.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Welding Qualification, Section IX

(Applications for copies should be addressed to the American Society of Mechanical Engineers, 345 East 4th Stree, New York, New York 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82	- Standard Practice for Commercial Packaging

(Applications for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish three shipping frames for ISO testing. The first article shall be manufactured using the same methods, procedures, and tooling proposed for production. Approval of the first article by the procuring activity shall not relieve the contractor of the obligation to supply equipment conforming to this specification. Changes or deviations in production units from the approved first article baseline shall require written authorization by the procuring activity.

3.2 Materials. Material used in fabrication of the shipping frame shall be as specified on Drawing No. 855001B0000. Materials shall be free of defects and imperfections that might affect the serviceability and appearance of the finished product. Substitute material shall not be utilized prior to written authorization by the procuring activity. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3 Design and construction. The shipping frame shall be fabricated and assembled in accordance with Drawing No. 855001B0000 and as specified herein.

3.3.1 Cargo netting tiedowns. One cargo net conforming to MIL-N-27444, type HCU-15/C and four cargo tiedowns conforming to MIL-T-27260, type CGU-1/B, 6,000 pound capacity shall be provided with each shipping frame.

3.3.2 Connectors. Three horizontal connectors and two vertical connectors shall be provided with each shipping frame.

3.3.3 Safety approval plate. Necessary actions as prescribed in 49CFR451 (see code of Federal Regulations) shall be accomplished to obtain approval of the containers by an appropriate Convention for Safe Containers (CSC) approval authority (see 49CFR451, Subpart B - "Approval of New Containers") and the required CSC safety approval plate shall be provided and affixed to the frame as prescribed in 49CFR451, C - "Safety Approval Plate". The notice of approval received from the approval authority which authorizes the attachment of CSC safety approval plates shall be furnished to the contracting officer within 30 days of receipt by the contractor. The contractor shall also provide a certificate to the contracting officer stating the serial number, model number, and any pertinent manufacturing identification numbers of the shipping frames that are covered under a specific notice of approval.

3.3.4 Welding and welders.

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3.3.4.1 Welding. The shipping frame shall be welded in accordance with Drawing No. 855001B0000. Welding shall meet the requirements of MIL-W-8611. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Spot, tack, or intermittent welds shall not be permitted unless otherwise cited in Drawing No. 855001B0000. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration. All finished weldments shall be straight and true in accordance with good shop practice. Faulty weldments shall be corrected by welding only. Peening shall not be utilized to correct faults. Weld spatters shall be removed. All parts shall be properly aligned and meshed and operable without interference, tight spots or other irregularities.

3.3.4.2 Welders. All welders shall be qualified for the types of welds required in the fabrication of shipping frames. Before assigning any welder to welding work, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operation to be performed and that such qualification is effective as defined by that particular code.

AWS D1.1, Structural Welding Code, Section 5.
Qualification - Welding Inspection Manual

Boiler and Pressure Vessel Code, Welding Qualifications, Section IX (ASME)

3.3.5 Castings and forgings. Castings and forgings shall be manufactured in accordance with Drawing No. 855001B0000. All castings shall be sound and free from patching, inclusions, misplaced coring, warping, or any other defect that reduces the castings' ability to perform its intended function. Forgings shall be uniform in quality and condition, and shall be free from tears, cracks, seams, laps, internal ruptures, imbedded scale, segregation and other defects.

3.3.6 Bolted and pinned connections. Bolt and pin holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice and all bolts, nuts and screws shall be tight.

3.3.7 Machine work. Where machined surfaces are required to be heat treated, hardened, finish grounded, plated or coated, the part dimensions, as specified in Drawing No. 855001B0000, shall be maintained.

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3.3.8 Metal fabrication. Metal used in the fabrication of the shipping frames shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Flame cutting, using a tip suitable for the thickness of the metal, may be employed instead of shearing or sawing. Reentrant cuts may be made with controlled means in order to ensure uniformity of size and shape. Heated metal shall be allowed to cool slowly.

3.3.9 Interchangeability. Shipping frames of the same classification, furnished under any specific contract, shall be physically and mechanically identical. They shall meet the applicable requirements specified in ISO 668, 1161, and 1496/I.

3.4 Environmental requirements. Shipping frames shall withstand, without cracking, corroding, peeling of paint, or loss of functionality, humidity resistance, low temperature (-25°F) and high temperature (160°F) when tested as specified in 4.5.9.

3.5 Painting. The shipping frame shall be painted as specified on Drawing No. 855001B0000 and MIL-E-52798. Color shall be as specified in 6.2.

3.6 Marking/serialization.

3.6.1 Shipping frame. The shipping frame shall be marked and serialized as specified on Drawing No. 855001B0000.

3.6.2 Parts and components. Parts and components shall be marked in accordance with MIL-STD-130.

3.6.3 Identification plate. The shipping frame identification plate shall be in accordance with Drawing No. 855001B0000.

3.7 Workmanship. All parts, components, and assemblies of the shipping frame, including castings, forging, molded parts, stampings, seals and sealing agents, machined surfaces, and welded parts shall be clean and free from any defects that will reduce the capability of the shipping frame to meet the requirements specified herein. Any components and assemblies which have been repaired or modified to overcome deficiencies shall not be used without prior approval of the contracting officer. External surfaces shall be free of burrs, slag, sharp edges and corners where sharp edges and corners are required.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirement.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)
(c) Packaging (see 4.6)

4.3 First article inspection and testing. First article inspection shall consist of the examinations and tests specified in Table I. Failure to pass any test shall be cause for rejection. Three SIXCON shipping frames shall be used for testing.

TABLE I. Inspection and test requirements

Inspection	Requirement Paragraph	Test Paragragh	First Article	Quality Conformance
Material Certification	3.2	4.5.1	X	X
Construction	3.3	4.5.2	X	X
Painting	3.5	4.5.3	X	X
Marking/Serialization	3.6	4.5.4	X	X
Workmanship	3.7	4.5.5	X	X
Welds	3.3.4	4.5.6	X	X
Castings	3.3.5	4.5.7	X	X
Forgings	3.3.5	4.5.8	X	X
Environmental Test	3.4	4.5.9	X	
ISO Testing	3.3.3	4.5.10	X	
Packaging	5	4.6	X	

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in Table I.

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4.4.1 Lot size. An inspection lot shall consist of all shipping frames produced during an identifiable production period and submitted at one time for acceptance. Production period shall be 1 month.

4.4.2 Sampling for inspection. Sampling for inspection from each inspection lot shall be in accordance with inspection N-II-S of MIL-STD-105. The Acceptance Quality Level (AQL) shall be 0.65 percent for critical defects, 1.0 percent for major defects and 2.5 percent for minor defects.

4.4.3 Inspection failure. Failure of a production shipping frame to meet any of the requirements specified herein during and as a result of the specified inspections and tests shall be cause for refusal by the Government to continue the acceptance of production shipping frames until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies in shipping frames previously accepted and produced under the contract shall be accomplished by the contractor at no cost to the Government. Any deficiencies found as a result of any production inspection will be considered prima facie evidence that all of any production inspection that all shipping frames accepted prior to the completion of the production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Examinations and tests.

4.5.1 Material certification. The contractor shall certify that all materials and parts used in fabrication of the shipping frames are in accordance with the requirements of 3.2. Material or part manufacturer's certification of test data will be acceptable for fulfilling this requirement. Acceptance or approval of any material or part shall not be construed as a guarantee of acceptance of the shipping frames. The contractor shall be responsible for using parts and assemblies from Qualified Products Lists QPL's) whenever available. Contractor's inspection records shall specifically list all QPL items by number and date of the QPL, name of contractor, and part number or drawing number(s). When parts and assemblies are approved as qualified products, but not yet listed on the QPL, the contractor shall list the products by number and date of the approved document, and name of supplier(s).

4.5.2 Construction. The shipping frame shall be visually examined both prior to and after painting to verify proper assembly and installation of components and assemblies in accordance with the requirements of 3.3.

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4.5.3 Painting. The shipping frame shall be inspected to verify conformance to the requirements of 3.5.

4.5.4 Marking/serialization. The shipping frame shall be visually examined to verify conformance to the requirements of 3.6.

4.5.5 Workmanship. The shipping frame shall be visually examined to verify conformance to the requirements of 3.7.

4.5.6 Welds. Welds shall be inspected in accordance with MIL-I-6866 or MIL-I-6868 to determine conformance to the requirements of 3.3.4.

4.5.7 Castings. Castings shall be inspected in accordance with MIL-STD-453 to determine conformance to the requirements of 3.3.5.

4.5.8 Forgings. Forgings shall be inspected in accordance with MIL-I-6868 to determine conformance to the requirements of 3.3.5.

4.5.9 Environmental tests.

4.5.9.1 Humidity resistance. The shipping frame under test shall be subjected to humidity resistance test in accordance with MIL-STD-810, method 507.1, procedure 4. After testing, the shipping frame shall be inspected for evidence of cracking, cracked welds, corrosion, and peeling of paint. Evidence of these conditions of material degradation shall constitute failure of the humidity test.

4.5.9.2 Low temperature. After having been tested and inspected in accordance with the provisions of 4.5.9.1, the shipping frame shall be soaked at -25°F for 16 hours minimum. The temperature shall then be raised to ambient conditions, and the shipping frame inspected for cracking, cracked welds, and paint disintegration. Evidence of these conditions or material degradation shall constitute failure of the low temperature test.

4.5.9.3 High temperature. After having been tested and inspected in accordance with 4.5.9.2, the shipping frame shall be soaked at 160°F (70°C) for 48 hours. The temperature shall then be lowered to ambient conditions, and the shipping frame inspected for cracking cracked welds, and paint disintegration. Evidence of these conditions or material degradation shall constitute failure of the high temperature test.

4.5.10 ISO testing. Three shipping frames shall be used for the complete series of ISO testing. ISO testing shall be carried out as prescribed in ISO 1496/I-1978(E) for half-high IC containers with a maximum gross weight of 30,000 pounds. All necessary ISO testing shall be accomplished to obtain Convention for Containers (CSC) safety approval plates (3.3.3).

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4.6 Packaging. The preservation, packaging, packing and container marking shall be inspected, prior to shipment, to determine conformance to section 5 of this specification.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or industrial as specified (see 6.2).

5.1.1 Level A. The shipping frame shall be preserved and packaged in accordance with the applicable procedures of MIL-P-116 and MIL-STD-794.

5.1.2 Level B. The shipping frame shall be preserved and packaged in accordance with the applicable procedures of MIL-P-116 and MIL-STD-794.

5.1.3 Industrial. The shipping frame shall be preserved and packaged in accordance with ASTM D3951-82.

5.2 Packing. Packing shall be level A, B, or industrial as specified (see 6.2).

5.2.1 Level A. The shipping frame shall be packed in accordance with the applicable procedures of MIL-STD-794.

5.2.2 Level B. The shipping frame shall be packed in accordance with the applicable procedures of MIL-STD-794.

5.2.3 Industrial. The shipping frame shall be packed in accordance with ASTM-D3951-82.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The shipping frames are intended to be used for mounting and transporting tanks, pumps, and other gear for use by combat troops in the field.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition requirements should specify the following:

(a) Title, number, and date of this specification.

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(b) When first article is not required (see 3.1).

(c) Color required (see 3.5).

(d) Levels of preservation, packaging, and packing (see 5.1 and 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a Contract Data Requirements List (CDRL) (DD Form 1423), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Para No.	Data Requirement Title.	Applicable DD No	Option
4.3	First Article Inspection Procedure	DI-T-4901	--
4.3	First Article Inspection Report	DI-T-4902	--

(Data item descriptions related to this specification and identified in section 6 will be approved and listed as such as DOD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.

6.3 First article. Any changes or deviations of production shipping frames from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the supplier of this obligation to furnish shipping frames conforming to this specification.

Preparing activity:
Navy - MC
Project No. 8145-N022

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-29419(MC)	Frame, Shipping (SIXCON)

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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